

moderation, and image search and organisation.

▪ **Amazon Lex**

Amazon Lex is a service for building conversational interfaces using voice and text, powered by the same deep learning technologies that underpin Amazon Alexa. Lex provides automatic speech recognition (ASR) to convert speech to text and natural language understanding (NLU) to recognise the intent of the text. It supports multi-turn conversations and context management, enabling the creation of sophisticated chatbots and voice assistants. Lex integrates seamlessly with AWS Lambda for executing business logic and can be connected to other AWS services to build comprehensive, interactive applications.

▪ **Amazon Polly**

Amazon Polly is a text-to-speech (TTS) service that converts text into lifelike speech using advanced deep learning technologies. Polly offers a wide range of voices and languages, providing high-quality speech synthesis with low latency suitable for real-time applications. It supports Speech Synthesis Markup Language (SSML) for fine-grained control over speech output, including pronunciation, volume, and speech rate adjustments. Polly is widely used in e-learning platforms, content accessibility tools, and voice-enabled devices to enhance user engagement through natural-sounding speech.

▪ **Amazon Comprehend**

Amazon Comprehend is a natural language processing (NLP) service that uses machine learning to extract insights from text. It can identify entities, perform sentiment analysis, detect languages, classify documents, and analyse relationships between entities. Comprehend supports custom entity recognition and topic modelling, allowing users to train models specific to their domain. It integrates with other AWS services for seamless data processing

and is utilised in customer feedback analysis, content categorisation, and knowledge discovery to drive business intelligence.

▪ **Amazon Transcribe**

Amazon Transcribe is an automatic speech recognition (ASR) service that converts spoken language into text. It provides real-time and batch transcription with features like speaker identification and custom vocabulary for domain-specific terms. Transcribe is designed to handle various audio formats and noisy environments, delivering high accuracy in transcription. It is used in industries such as contact centres for call analysis, media for generating subtitles, and legal and healthcare sectors for documentation purposes, enhancing efficiency and accuracy in transcription workflows.

▪ **Amazon Translate**

Amazon Translate is a neural machine translation service that delivers fast, high-quality translations across numerous languages. It uses advanced deep learning models to provide accurate and fluent translations, supporting real-time and batch processing. Translate allows customisation with custom terminology, ensuring consistency in translating domain-specific terms. It integrates with other AWS services to enable seamless localisation of applications, multilingual customer support, and global communication, helping businesses expand their reach by overcoming language barriers.

▪ **Amazon Textract**

Amazon Textract is a machine learning service that automatically extracts text, handwriting, and data from scanned documents. It goes beyond simple OCR by identifying the structure of documents, such as forms and tables, to extract data accurately. Textract supports integration with other AWS services

for automated processing and analysis of extracted data. It is used in digitising paper records, automating data entry, and ensuring regulatory compliance through accurate extraction and analysis of information from documents, streamlining business operations.

▪ **Amazon Kendra**

Amazon Kendra is an intelligent search service that uses machine learning to deliver highly accurate and relevant search results. It understands natural language queries and indexes content from various data sources, providing contextual and relevant information. Kendra allows customisation of search relevance based on user feedback and supports features like document ranking and faceted search. It is utilised in enterprise search applications, enhancing knowledge management systems, customer support portals, and any scenario requiring efficient and accurate information retrieval.

GCP AI services

Google Cloud Platform (GCP) offers a wide range of AI services designed to help developers and businesses incorporate machine learning and artificial intelligence into their applications.

▪ **Cloud AI Platform**

This is a comprehensive suite of tools for building, deploying, and managing machine learning models at scale. It comprises:

AI platform notebooks: These manage Jupyter Notebooks for ML development.

AI platform training: Scalable training of ML models using various frameworks.

AI platform prediction: Scalable serving of ML models for online predictions.

AI platform pipelines: These orchestrate ML workflows using Kubeflow pipelines.

▪ **AutoML**

The tools under this service enable